Why the Gutzwiller wave function Ψ_G is not normalized

(Hubbard dimer; worked example: $N_g = 0$, $\Delta V = 0$)

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The wave function $|\Psi\rangle = P_0 P_1 |\Psi_0\rangle$ built from a converged gGA solution (Sec. "The Gutzwiller wave function itself" of `results.tex`) has $\langle\Psi|\Psi\rangle \neq 1$ for the dimer. This note shows where the norm comes from, which part of it the gGA constraints fix, which part they cannot, and evaluates it in closed form for $N_g = 0$, $\Delta V = 0$, checked against the computed values (script `psi_norm_check.py`, output `psi_scan/psi_norm_check.txt`).

1 The norm as the expectation value of a product of two local operators

For fragment $i = 0, 1$ let $|\Psi_0\rangle$ be the Slater determinant of the occupied levels of H_{qp}, with qp Fock space $\{|n\rangle\}$ ($4^{N_{\rm eff}}$ states per fragment, $N_{\rm eff} = 1 + N_g$), and let $P_i^0 = \langle\Psi_0|\,|n\rangle\langle n'|\,|\Psi_0\rangle$ be its local density matrix on fragment i. The projector is $P_i = \sum_{\Gamma n} \lambda_{i,\Gamma n} |\Gamma\rangle\langle n|$, with $\lambda_i = \tilde{\Phi}_i \, (P_i^0)^{-1/2}$ and $\tilde{\Phi}_i[\Gamma, n]$ the embedding state of the fragment, $\sum_{\Gamma n} \tilde{\Phi}_{i,\Gamma n}^2 = \langle\phi_i|\phi_i\rangle = 1$ (all real). Define the operator on the qp Fock space of fragment i

$$G_i \equiv P_i^\dagger P_i = \lambda_i^T \lambda_i = (P_i^0)^{-1/2} \, \tilde{\Phi}_i^T \tilde{\Phi}_i \, (P_i^0)^{-1/2}. \tag{1}$$

Since P_0 and P_1 act on different fragments, $\langle\Psi|\Psi\rangle = \langle\Psi_0|P_1^\dagger P_0^\dagger P_0 P_1|\Psi_0\rangle$ is

$$\langle\Psi|\Psi\rangle = \langle\Psi_0|\, G_0 \, G_1 \,|\Psi_0\rangle. \tag{2}$$

2 What the gGA constraints fix: each factor alone

The expectation value of a single factor in Ψ_0 is a trace with the local density matrix, and with Eq. (1)

$$\langle\Psi_0|G_i|\Psi_0\rangle = \mathrm{Tr}\big[P_i^0 \, G_i\big] = \mathrm{Tr}\big[\tilde{\Phi}_i^T \tilde{\Phi}_i\big] = \langle\phi_i|\phi_i\rangle = 1, \tag{3}$$

up to the weight of $\tilde{\Phi}_i$ outside the support of P_i^0 (at most 7×10^{-8} in the computed points). This is the normalization condition of the embedding state. The density constraint $\Delta_{ii}^* = 1 - \Delta_{\phi_i}$ is the corresponding statement for the one-body operators $f_a^\dagger f_b$ of the fragment: in their expectation values G_i may be replaced by 1 (the standard form of the Gutzwiller constraints; this part is not checked separately here). Both are statements about one fragment. They do not say that G_i is the identity operator: G_i is a nontrivial operator whose expectation value, and whose one-body moments, are those of the identity.

3 What they cannot fix: the correlation of G_0 and G_1

Writing the product in Eq. (2) in terms of the single-fragment values and a connected part,

$$\langle\Psi|\Psi\rangle = \langle G_0\rangle\langle G_1\rangle + \Big(\langle G_0 G_1\rangle - \langle G_0\rangle\langle G_1\rangle\Big) = 1 + \Big(\langle G_0 G_1\rangle - \langle G_0\rangle\langle G_1\rangle\Big), \tag{4}$$

where all expectation values are in $|\Psi_0\rangle$ and Eq. (3) was used. The second term is the correlation, in Ψ_0, between the two fragment operators G_0 and G_1. In a dimer, Ψ_0 is a Slater determinant of two sites with delocalized orbitals, and the two fragments are correlated: this term is not zero. The gGA is the Gutzwiller approximation, where the factorization $\langle G_0 G_1\rangle \to \langle G_0\rangle\langle G_1\rangle$ is taken to hold (it is the limit in which the connected term vanishes); the constraints of Sec. 2 are exactly what makes the norm equal to 1 then. For the dimer the factorization is not exact, and $\langle\Psi|\Psi\rangle - 1$ is the connected term of Eq. (4). It is therefore not a numerical error of the construction: for the converged points $\langle G_0\rangle = \langle G_1\rangle = 1$ holds to 4×10^{-9} and the connected term carries all of the deviation (Table 1).

4 Example: $N_g = 0$, $\Delta V = 0$ in closed form

Here $N_{\text{eff}} = 1$ and everything can be written out.

Ψ_0. $H_{qp} = -tZ\sigma_x$ and the occupied orbital is the bonding orbital $b = (c_0 + c_1)/\sqrt{2}$, so

$$|\Psi_0\rangle = b_\uparrow^\dagger b_\downarrow^\dagger|0\rangle = \tfrac{1}{2}\big(c_{0\uparrow}^\dagger + c_{1\uparrow}^\dagger\big)\big(c_{0\downarrow}^\dagger + c_{1\downarrow}^\dagger\big)|0\rangle, \tag{5}$$

which is a superposition of four configurations of equal weight $\frac{1}{4}$: $(\uparrow\downarrow, 0)$, $(0, \uparrow\downarrow)$, $(\uparrow, \downarrow)$ and $(\downarrow, \uparrow)$ (state of site 0, state of site 1). Each fragment has the occupations $n_\sigma = \frac{1}{2}$ and no local correlation, so $P_i^0 = \frac{1}{4}\mathbb{1}$ on its four configurations $0, \uparrow, \downarrow, \uparrow\downarrow$.

λ_i and G_i. With $d \equiv \langle n_\uparrow n_\downarrow\rangle = \frac{1}{4} - \frac{U}{32t}$ (the closed Gutzwiller solution of Sec. "The $N_g = 0$ case" of `results.tex`), the embedding state has the weights d on the empty and on the doubly occupied configuration (particle–hole symmetry at $\Delta V = 0$) and $\frac{1}{2} - d$ on each singly occupied one. The embedding state pairs each physical configuration with one qp configuration, so $\tilde{\Phi}_i$ and hence $\lambda_i = \tilde{\Phi}_i (P_i^0)^{-1/2} = 2\tilde{\Phi}_i$ are diagonal and $G_i = \lambda_i^T \lambda_i$ is the diagonal operator with entries $4\times$(weight of the configuration):

$$G_i = 4d \quad \text{on } |0\rangle,\ |\uparrow\downarrow\rangle, \qquad G_i = 2 - 4d \quad \text{on } |\uparrow\rangle,\ |\downarrow\rangle. \tag{6}$$

Check of Eq. (3): $\langle G_i\rangle = \frac{1}{4}\left(2\cdot 4d + 2\cdot(2 - 4d)\right) = 1$.

The product. Let e_i be the indicator of even occupation of fragment i (1 on $|0\rangle, |\uparrow\downarrow\rangle$, 0 on the singly occupied ones). Then

$$G_i = (2 - 4d) + (8d - 2)\, e_i. \tag{7}$$

In Ψ_0: $\langle e_i\rangle = \frac{1}{2}$ (each fragment is even in two of the four configurations), but $\langle e_0 e_1\rangle = \frac{1}{2}$, since the configurations $(\uparrow\downarrow, 0)$ and $(0, \uparrow\downarrow)$ have both fragments even and $(\uparrow, \downarrow)$, $(\downarrow, \uparrow)$ both odd; if the fragments were uncorrelated it would be $\frac{1}{4}$. The connected part is $\langle e_0 e_1\rangle - \langle e_0\rangle\langle e_1\rangle = \frac{1}{4}$, and Eq. (4) gives

$$\langle\Psi|\Psi\rangle = 1 + (8d - 2)^2 \cdot \tfrac{1}{4} = 1 + (4d - 1)^2 = 1 + \Big(\frac{U}{8t}\Big)^2, \tag{8}$$

using $4d - 1 = -U/8t$. It is 1 at $U = 0$ (where $G_i = 1$) and grows to 2 at $U_c = 8t$, where $d = 0$ and G_i projects onto the two singly occupied configurations ($Z = 0$).

Comparison with the computed values. Table 1 gives the quantities of Eqs. (3)–(4) computed from the stored solutions with `build_psi`, and for $N_g = 0$ the closed form (8). They agree to 3×10^{-8} (the shift 10^{-9} in S^{-1}). The $N_g = 0$ values equal the $\langle\Psi|\Psi\rangle$ of the norm tables of `results.tex` (1.062, 1.250, 1.562, 1.766 at $U/t = 2, 4, 6, 7$).

N_g	U/t	$\langle G_0\rangle$	$\langle G_1\rangle$	$\langle G_0 G_1\rangle = \langle\Psi\|\Psi\rangle$	connected	$1 + (U/8t)^2$
0	2	1.0000000	1.0000000	1.0625000	0.0625000	1.0625000
0	4	1.0000000	1.0000000	1.2500000	0.2500000	1.2500000
0	6	1.0000000	1.0000000	1.5625000	0.5625000	1.5625000
0	7	1.0000000	1.0000000	1.7656250	0.7656250	1.7656250
2	2	1.0000000	1.0000000	1.1527890	0.1527890	–
2	4	1.0000000	1.0000000	1.7579779	0.7579779	–
2	6	1.0000000	1.0000000	2.6468041	1.6468041	–

Table 1: $\Delta V = 0$: single-fragment norms, norm of Ψ and connected part of Eq. (4), from the stored solutions. The $N_g = 2$ values are the $\langle\Psi|\Psi\rangle$ of `results.tex`; for $N_g = 2$ there is no closed form here.

5 Consequences

Since the connected term is a property of Ψ_0 and of the G_i, it is not removed by changing the solution of the gGA equations; the wave function is normalized by hand, $\Psi \to \Psi/\sqrt{\langle\Psi|\Psi\rangle}$, before any expectation value is computed (energy, overlap with FCI, P_{ion}, amplitudes), as in `results.tex`. For $N_g = 2$ the same structure holds, Eqs. (1)–(4), with $\langle G_i\rangle = 1$ and a connected term that grows with U/t (Table 1); I have not derived a closed form for it.